State-Dependent GRAPE: Gradient reformulation for PHIP-SAH polarization transfer in the presence of dipolar field

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Motivation

- **Parahydrogen Included Polarization via Side Arm Hydrogenation (PHIP-SAH)** transfers polarization from singlet-order parahydrogen to target ^{13}C nuclei can lead to a **dipolar field build-up over the sample volume, which reduces transfer efficiency and becomes a major limitation for achieving high polarization.**
- Classically engineered pulse sequences can provide strong robustness against B_0, B_1 inhomogeneities and chemical shift variations. However, their performance deteriorates in the presence of emerging dipolar fields.
- Achieving a pulse sequence that is **simultaneously robust against all relevant perturbations—namely B_0 and B_1 inhomogeneities, chemical shift variations, and dipolar field interaction**—remains a significant challenge and an active area of research in this field.
- The dipolar-field effect is fundamentally different from conventional imperfections such as B_0 and B_1 field inhomogeneities or chemical shift dispersion. While these effects can be treated as state-independent Hamiltonian terms, the **dipolar field depends on the instantaneous spin state, leading to state-dependent spin dynamics**.

These practical constraints motivate the search for a pulse sequence that is fully robust against all relevant non-ideal parameters.

Classical GRAPE algorithm

Assumption: State-independent Hamiltonian

System dynamics:

$$\dot{\rho}(t) = -i\left[\left(\mathcal{H}_0 + \sum_{k=1}^{m} u_k(t)\mathcal{H}_k\right), \quad \rho(t)\right].$$

GRAPE discrete propagator:

$$U_j = e^{-i\Delta t(\mathcal{H}_0 + \sum_k u_j \mathcal{H}_k)}$$

Forward evolution :

$$\rho(T) = U_N \cdots U_1 \rho_0 U_1^\dagger \cdots U_N^\dagger.$$

Performance index for target state C :

$$\phi \quad = \quad \langle C|\phi(T)\rangle \quad = \quad \langle \lambda_j|\rho_j\rangle,$$

$$\rho_j = U_j \cdots U_1 \rho_0 U_1^\dagger \cdots U_j^\dagger, \quad \lambda_j = U_{j+1}^\dagger \cdots U_1^\dagger \rho_0 U_1 \cdots U_{j+1}$$

Gradient derivation:

$$\frac{\partial \phi}{\partial u_{j+1}} = \langle \lambda_{j-1}|\frac{\partial U_{j+1}}{\partial u_{j+1}}\rho_j U_{j+1}^\dagger\rangle + \langle \lambda_{j-1}|U_{j+1}\rho_j \frac{\partial U_{j+1}^\dagger}{\partial u_{j+1}}\rangle,$$

$$\frac{\partial U_{j+1}}{\partial u_{j+1}} = (-i\Delta t) U_{j+1}\mathcal{H}_k$$

$$\boxed{\left(\frac{\partial \phi}{\partial u_{j+1}}\right)_k = i\Delta t\langle \lambda_j|[\rho_j, \mathcal{H}_k]\rangle.}$$

Drawbacks

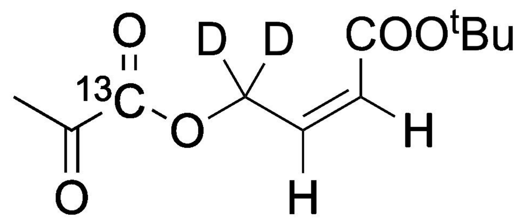
- Classical GRAPE optimization assumes a **state-independent** Hamiltonian during the forward and backward propagation. Therefore, in the presence of state-dependent dipolar-field dynamics, the standard gradient expression becomes insufficient, **leading to premature convergence and failing to achieve true fidelity**.

Key Contribution

- **Reformulation of the gradient of the GRAPE algorithm** by consistently **incorporating the state-dependent dipolar-field Hamiltonian** into the state and costate evolution, enabling optimal control design for nonlinear spin dynamics.
- The proposed modified GRAPE algorithm **improves convergence numerically** and **avoids premature optimization stagnation** without adding penalty terms to the cost function, demonstrating better performance than classical GRAPE for three-spin systems with dipolar interactions and other system imperfections.

System Model

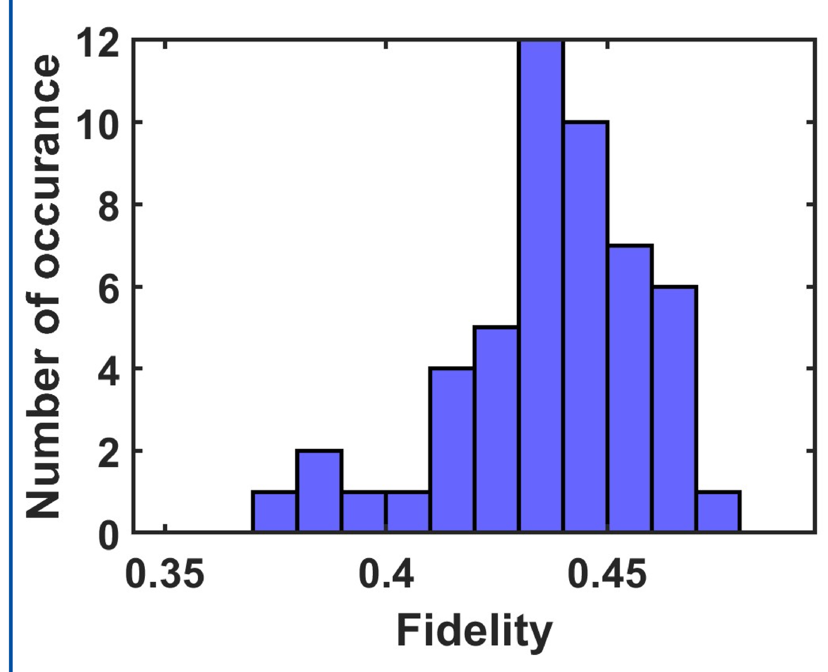
tert-butyl (Z)-4-((2-oxopropanoyl-1-^{13}C)oxy)but-2-enoate-4,4-d$_2$



- Three-spin system, $J_{12} = 11.7$ Hz, $J_{13} = 0.4$ Hz, $J_{23} = 0$ Hz
- 99% transfer time 3.46 sec

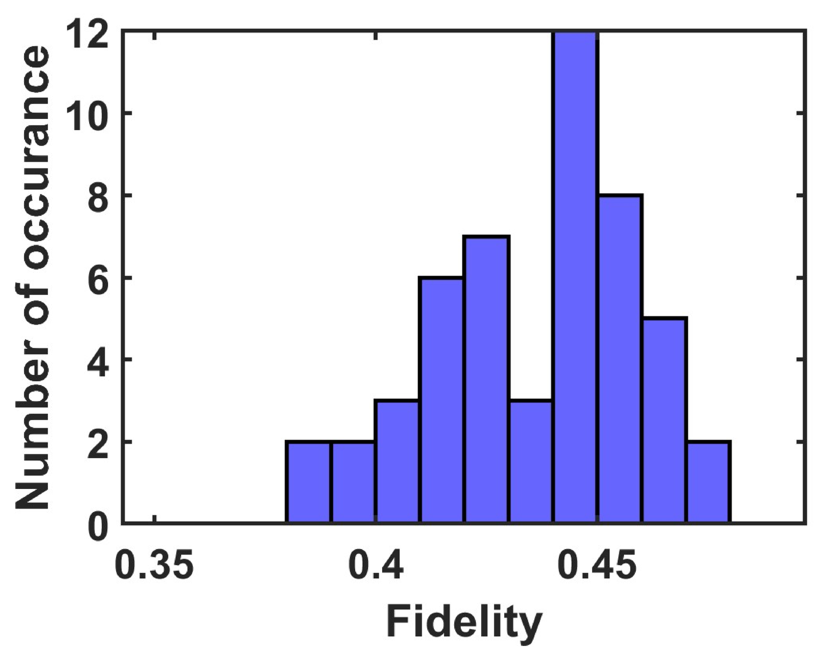
Convergence Analysis : Histogram Plot (optimization for 50 seeds)

Simulation parameter: $J_{12} = 11.7$ Hz, $J_{13} = 2$ Hz, reduced transfer time 0.641 sec, Control only on ^{13}C



Classical GRAPE with state-independent gradient

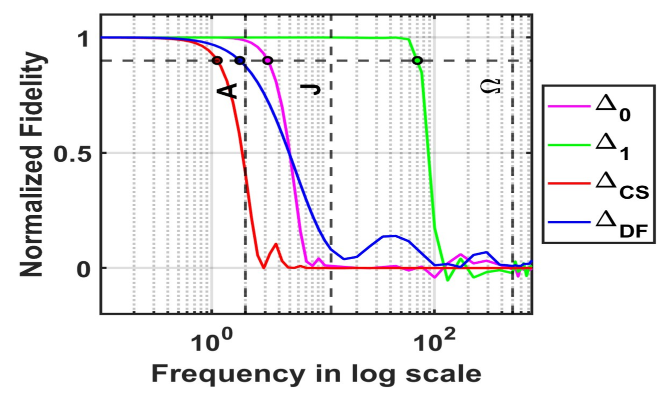
- Fidelity ranges between 0.45 and 0.5 is less, **Success rate** 28%.
- RMS deviation from 0.5 is 0.039905.
- Maximum fidelity 0.470554.
- Standard deviation within range 0.45 to 0.5 is 0.006183, **better.**
- **Better in terms of full distribution**.



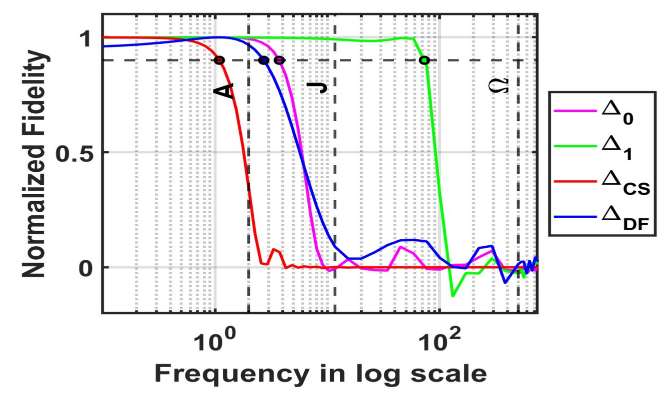
Dynamic GRAPE with state-dependent gradient

- Fidelity ranges between 0.45 and 0.5 is more, **Success rate** 30%.
- RMS deviation from 0.5 is 0.039747, **better.**
- Maximum fidelity 0.474605, **better.**
- Standard deviation within range 0.45 to 0.5 is 0.007424.
- **Better high-fidelity performance.**

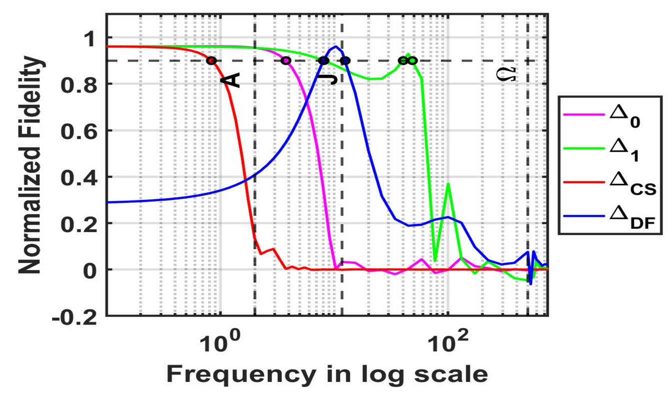
Optimization with Dipolar Field : Robustness Benchmark Plot (with Dynamic GRAPE algorithm)



$\Delta B_0 = [-1, 1]$ Hz, $\Delta B_1 = [-0.1, 0.1]\%$, $\Delta B_{Df} = 0$ Hz



$\Delta B_0 = [-1, 1]$ Hz, $\Delta B_1 = [-0.1, 0.1]\%$, $\Delta B_{Df} = 1$ Hz

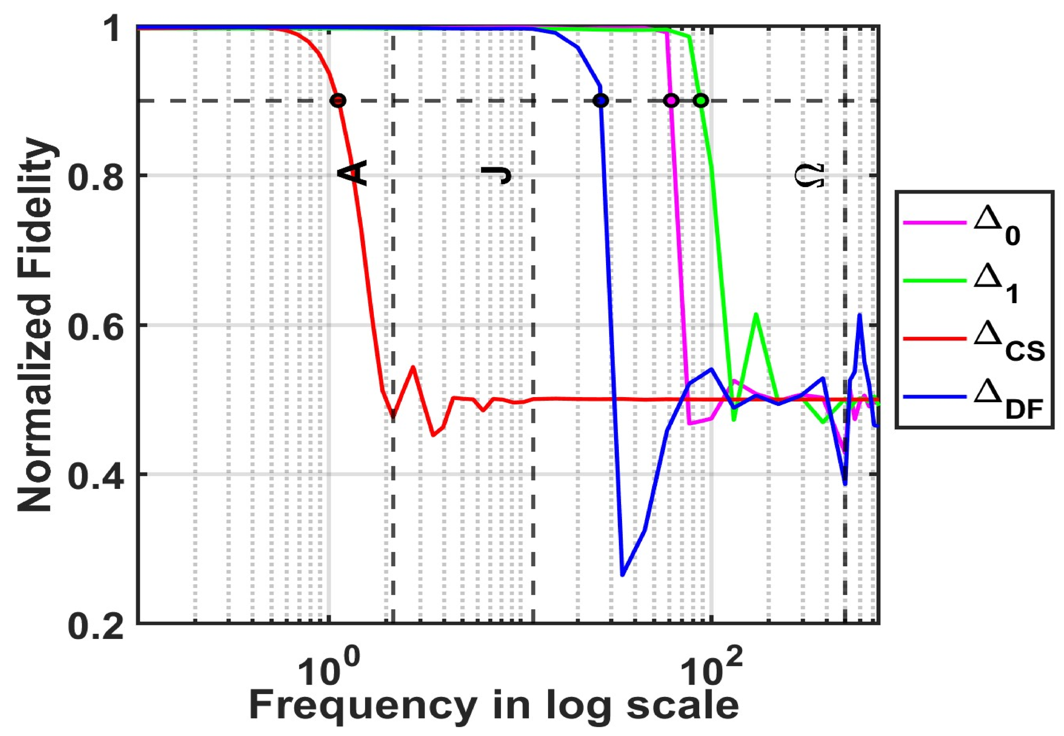


$\Delta B_0 = [-1, 1]$ Hz, $\Delta B_1 = [-0.1, 0.1]\%$, $\Delta B_{Df} = 10$ Hz

Table 1: Robustness benchmark values for parameter $\Delta B_0 = [-1, 1]Hz$, $\Delta B_1 = [-0.1, 0.1]$ Hz

Parameters	$\Delta B_{Df} = 0$ Hz	$\Delta B_{Df} = 1$ Hz	$\Delta B_{Df} = 10$ Hz
Δ_0	3.166	3.72	3.76
Δ_1	69.91	73.06	48.49
Δ_{CS}	1.11	1.09	0.834
Δ_{DF}	1.78	2.71	8.03 - 12.46

Optimization parameters $\Delta B_0 = [-60, 60]Hz$, $\Delta B_1 = [-0.1, 0.1]\%$, $\Delta B_{Df} = [0, 10]$ Hz, $\Delta B_{CS} = [-0.5, 0.5]$ Hz, Optimum Pulse time 1.2 sec, $J_{12} = 11.7$ Hz, $J_{13} = 2.1667$ Hz



- **The robustness benchmark values for this plot $\Delta_0 = 61.62$ Hz, $\Delta_1 = 87.984$ Hz, $\Delta_{CS} = 1.117$ Hz, $\Delta_{DF} = 26.397$ Hz (For normalized fidelity threshold value 0.9)**
- **Normalized fidelity reached upon optimization is 0.996.**

Classical vs Dynamic GRAPE

Aspect	Classical GRAPE	Dynamic GRAPE (This Work)
Hamiltonian	**State-independent**	**State-dependent** including dipolar field
Forward evolution	Uses **fixed propagators**	**State-dependent** propagators
Backward evolution	**State-independent** evolution	**State-dependent**
Gradient expression	$i\Delta t[\rho_j, \mathcal{H}_k]$	$i\Delta t[\rho_j, \mathcal{H}_k] - [\rho_j, \partial \mathcal{H}^{df}/\partial u]$
Convergence behavior	**Premature** under nonlinearities	**Stable**, fidelity-driven

Dynamic GRAPE: Key Idea

Modified system equation:

$$\dot{\rho}(t) = -i\left[\left(\mathcal{H}_0 + \mathcal{H}^{df} + \sum_k u_k(t)\mathcal{H}_k\right), \quad \rho(t)\right].$$

Propagator with dipolar term:

$$U_{j+1} = e^{-i\Delta t(\mathcal{H}_0 + \mathcal{H}_{j+1}^{df} + \sum_k u_{j+1}\mathcal{H}_k)}.$$

Forward evolution :

$$\rho(T) = U_N \cdots U_1 \rho_0 U_1^\dagger \cdots U_N^\dagger.$$

Performance index for target state C :

$$\phi \quad = \quad \langle C|\phi(T)\rangle \quad = \quad \langle \lambda_j|\rho_j\rangle,$$

$$\rho_j = U_j \cdots U_1 \rho_0 U_1^\dagger \cdots U_j^\dagger, \qquad \lambda_j = U_{j+1}^\dagger \cdots U_1^\dagger \rho_0 U_1 \cdots U_{j+1}$$

Gradient with time-varying $\mathcal{H}^{df}$:

$$\frac{\partial U_{j+1}}{\partial u_{j+1}} = (-i\Delta t)U_{j+1}\left(\mathcal{H}_k + \frac{\partial \mathcal{H}_{j+1}^{df}}{\partial u_{j+1}}\right),$$

$$\boxed{\left(\frac{\partial \phi}{\partial u_{j+1}}\right)_k = i\Delta t\langle \lambda_j|[\rho_j, \mathcal{H}_k]\rangle - \langle \lambda_j|[\rho_j, \frac{\partial \mathcal{H}_{j+1}^{df}}{\partial u_{j+1}}]\rangle.}$$

Dipolar-field Hamiltonian (spin 3):

$$\mathcal{H}_{j+1}^{df} = 2\pi[2\langle I_{3z}\rangle \mathbb{I}_{3z} - \langle I_{3x}\rangle \mathbb{I}_{3x} - \langle I_{3y}\rangle \mathbb{I}_{3y}].$$

Derivative of dipolar Hamiltonian:

$$\frac{\partial \mathcal{H}_{j+1}^{df}}{\partial u_{j+1}} = 2\pi\left[2\frac{\partial\langle I_{3z}\rangle}{\partial u_{j+1}}\mathbb{I}_{3z} - \frac{\partial\langle I_{3x}\rangle}{\partial u_{j+1}}\mathbb{I}_{3x} - \frac{\partial\langle I_{3y}\rangle}{\partial u_{j+1}}\mathbb{I}_{3y}\right].$$

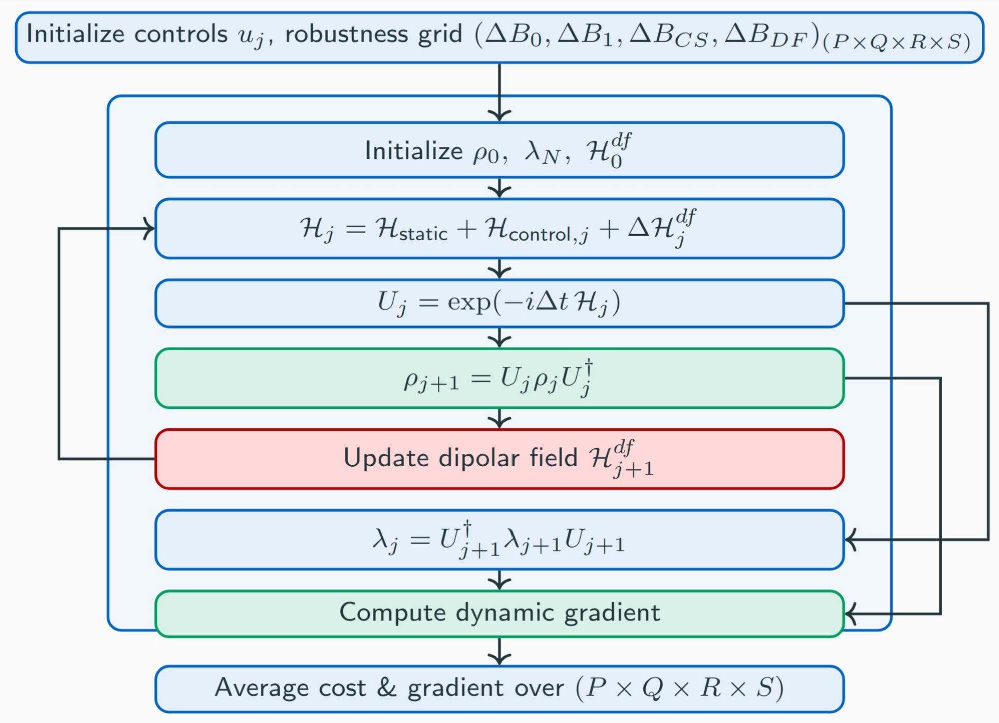
where,

$$\frac{\partial\langle I_{3z}\rangle}{\partial u_{j+1}} = \mathrm{Tr}\{i\Delta t[\rho_{j+1}, \mathcal{H}_k]\mathbb{I}_{3z}\}.$$

Hence,

$$\boxed{\frac{\partial \mathcal{H}_{j+1}^{df}}{\partial u_{j+1}} = 2\pi i\Delta t\Big[2\,\mathrm{Tr}\{[\rho_{j+1}, \mathcal{H}_k]\mathbb{I}_{3z}\}\mathbb{I}_{3z} - \mathrm{Tr}\{[\rho_{j+1}, \mathcal{H}_k]\mathbb{I}_{3x}\}\mathbb{I}_{3x} - \mathrm{Tr}\{[\rho_{j+1}, \mathcal{H}_k]\mathbb{I}_{3y}\}\mathbb{I}_{3y}\Big].}$$

Algorithmic Flow



Conclusions

- The **state-dependent gradient formulation** explicitly accounts for the dipolar fields generated by the target ^{13}C nuclei during polarization transfer.
- **Dynamic GRAPE improves the stability of the optimization process and enables more reliable convergence toward high-fidelity pulse design.**
- The proposed approach enables **robust PHIP-SAH polarization-transfer pulse design** by simultaneously compensating for dipolar-field effects and other practical system imperfections.

References

- Dagys, Laurynas, Martin C. Korzeczek, Anna J. Parker, James Eills, John W. Blanchard, Christian Bengs, Malcolm H. Levitt, Stephan Knecht, Ilai Schwartz, and Martin B. Plenio. "Robust parahydrogen-induced polarization at high concentrations." Science Advances 10, no. 30 (2024): eado0373.
- Korzeczek, Martin C., Laurynas Dagys, Christoph Müller, Benedikt Tratzmiller, Alon Salhov, Tim Eichhorn, Jochen Scheuer, Stephan Knecht, Martin B. Plenio, and Ilai Schwartz. "Towards a unified picture of polarization transfer—pulsed DNP and chemically equivalent PHIP." Journal of Magnetic Resonance 362 (2024): 107671.
- Korzeczek, Martin C., Ilai Schwartz, and Martin B. Plenio. "PHIP sequences and dipolar fields I-single spin control." Journal of Magnetic Resonance (2026): 108058.
- Khaneja, Navin, Timo Reiss, Cindie Kehlet, Thomas Schulte-Herbrüggen, and Steffen J. Glaser. "Optimal control of coupled spin dynamics: design of NMR pulse sequences by gradient ascent algorithms." Journal of magnetic resonance 172, no. 2 (2005): 296-305.

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